Predicting All-NBA Selection

Luxman Maheswaran

Duke University

Dept. of Statistical Science

All-NBA Introduction

- All-NBA Selection
  - 15 Players
  - 1st, 2nd , 3rd Team
- Importance
  - Legacy
  - Money
- Decided by 100 experts
- Recent Rule Changes
  - 2023-2024 Onwards

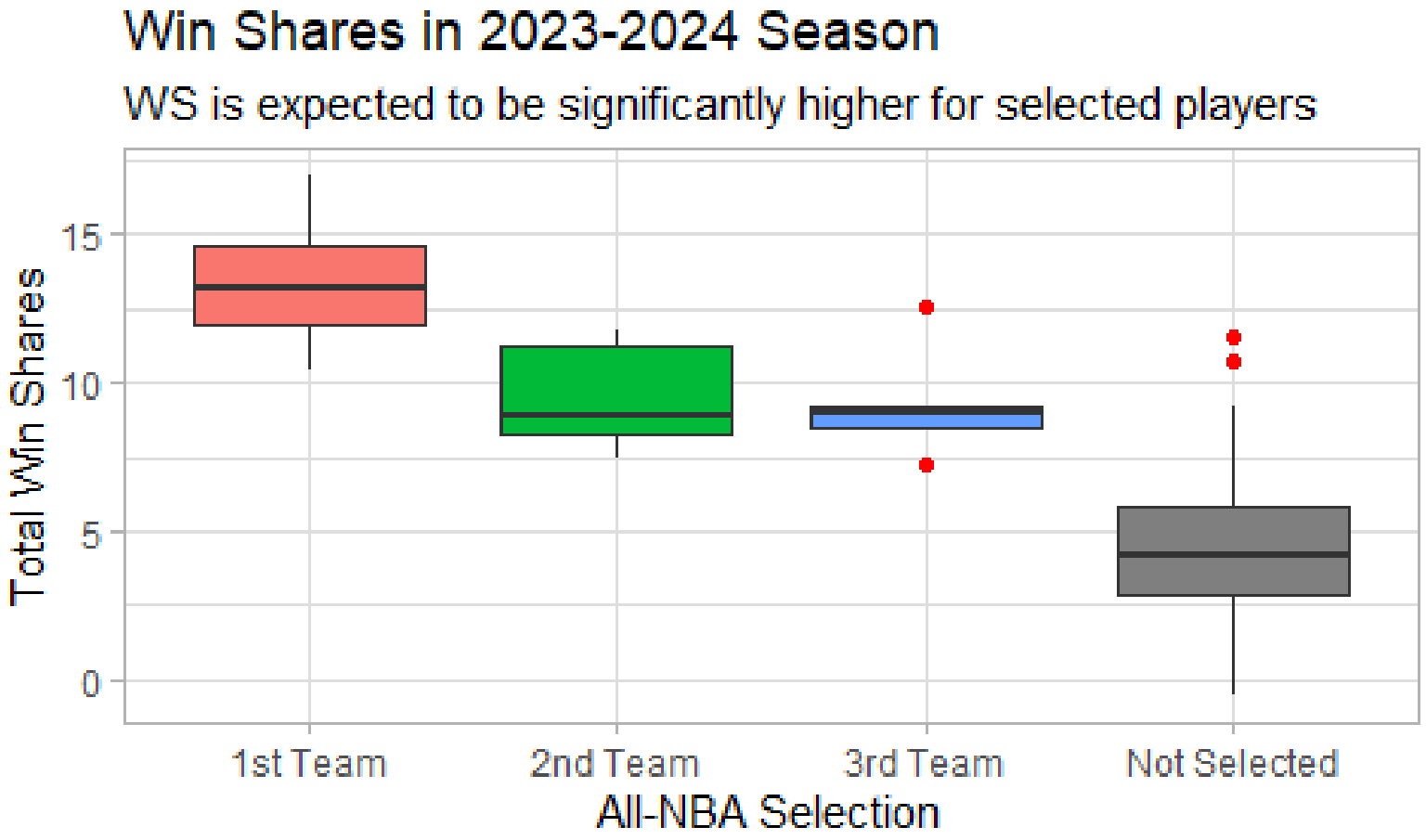
Data

- Basketball Reference
- 2 Complete Seasons
- Advanced Metrics
  - Win Share
    - Individual Player's Contribution
  - Box-plus-minus (BPM)
  - Value-over-replacement-player (VORP)

Player	MIN	PTS	FG	3PT	FT	REB	AST	STL	BLK	TO	PF	+/-
3 Jaden McDaniels · F	31	11	5-6	1-1	0-0	2	3	0	0	3	2	-40
30 Julius Randle · F	28	11	3-10	0-1	5-6	6	4	1	0	4	3	-12
27 Rudy Gobert · C	26	9	3-4	0-0	3-3	5	4	0	0	1	3	-38
5 Anthony Edwards · G	35	36	11-17	3-8	11-12	2	5	1	0	5	1	-25
0 Donte DiVincenzo · G	25	6	2-5	1-2	1-2	4	2	1	0	3	2	-12

- Goal
  - Predicting All-NBA Selections w/ Advanced Metrics

Exploratory Data Analysis



Modelling

- Binary Outcome
- Bayesian Logistic Regression
  - Small sample size
  - Sparse signal
- 14 Predictors
  - Avoid Complete Identification
  - Win Share
  - G, MP
  - FG, FGA, FT, FTA, PTS
  - TRB, AST, STL
  - TO, PF

$$logit(p_i) = \beta * X_i$$

i is a player
p_i is the probability of All-NBA selection
β is a vector of *j* coefficients
X_i is the design matrix

Prior Selection

- Horseshoe Prior
  - Shrinkage
  - Carvalho (2009)

$$\beta_j \mid \tau^2, \lambda_j^2 \sim N(0, \tau^2 \lambda_j^2)$$

$$\lambda_j \sim C^+(0,1)$$

- Hyperprior for τ
  - Piironen (2017)
  - Expected number of non-zero predictors
  - 5/14
  - $\frac{p_0}{D-p_0} \sim 1/2$

$$\tau \sim C^+(0, \tau_0)$$

Posterior Estimates for All-NBA Predictors

	Estimate	Est.Error	Q2.5	Q97.5
Intercept	-7.62	2.71	-14.69	-4.12
ratioWS	2.87	1.41	0.51	6.20
G	-0.32	0.61	-1.97	0.47
MP	-0.30	0.83	-2.66	0.88
FG	0.95	1.30	-0.53	4.11
FGA	0.46	1.03	-0.93	3.13
FT	-0.02	0.63	-1.45	1.28
FTA	-0.29	0.79	-2.41	0.72
TRB	0.07	0.51	-0.92	1.36
AST	0.25	0.55	-0.56	1.68
STL	-0.02	0.35	-0.85	0.72
BLK	-0.05	0.37	-0.94	0.64
TOV	0.81	0.96	-0.25	3.08
PF	-0.58	0.69	-2.27	0.23
PTS	0.98	1.45	-0.66	4.64

Discussion

- Diagnostics
- Similar Coef. For BPM, VORP
- Validation on 2024-2025 Season
  - 80% (12/15) correct
  - Good Order of All-NBA Teams

Predicted All-NBA Selections

Player	All-NBA	Predicted Label	Estimate	Estimate Uncertainty
Nikola Jokic	1st Team	1	1.00	0.01
Shai Gilgeous-Alexander	1st Team	1	1.00	0.01
Giannis Antetokounmpo	1st Team	1	0.96	0.09
Jayson Tatum	1st Team	1	0.85	0.13
Anthony Edwards	2nd Team	1	0.81	0.17
LeBron James	2nd Team	1	0.77	0.21
Stephen Curry	2nd Team	1	0.63	0.22
James Harden	3rd Team	1	0.60	0.31
Tyler Herro	Not Selected	1	0.59	0.26
Tyrese Haliburton	3rd Team	1	0.59	0.28
Karl-Anthony Towns	3rd Team	1	0.56	0.30
Ivica Zubac	Not Selected	1	0.55	0.30
Cade Cunningham	3rd Team	1	0.51	0.31
Jalen Brunson	2nd Team	1	0.50	0.24
Trae Young	Not Selected	1	0.45	0.34
Donovan Mitchell	1st Team	0	0.32	0.19
Evan Mobley	2nd Team	0	0.21	0.16
Jalen Williams	3rd Team	0	0.20	0.14

Limitations

- Small Dataset
  - Scraping Limits
  - Position Constraints
- Ignores Time Series Correlation
  - Past season performance
- Limited scope of box score

Future Directions

- Historical All-NBA Prediction
- Player Rankings
- Choice of Predictors/Model Design
- Betting Odds
  - Evolving probabilities

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AI Disclosure

- AI was used to create/edit/debug code, generate LaTex code, generate modelling ideas, and compare/contrast different ideas.

Thanks for Listening

Questions?